OBJECT ORIENTED PROGRAMMING
SECOND YEAR SECOND SEMESTER
ACTIVITY #2 - INTRODUCTION TO JAVA

Gomez, Dean Benedict B. **3/2/2026**
BSIT 2-1 **Prof. Bedis Jr., Severino**

PART I:

1. Create a Java program using bitwise operators that will produce the output: true, 10, false, 15.

CODE

Act2_Part1.java > Act2_Part1 > main(String[])

```java
public class Act2_Part1 {
    Run | Debug
    public static void main(String[] args) {
        int num1 = 10;
        int num2 = 15;

        boolean a = (num1 & num2) != 0;  // 10 & 15 is 10 which is != 0, result true
        int b = num1 & num2;             // 10 & 15 is 10, result: 10
        boolean c = (num1 | num2) != 15; // 10 | 15 is 15 which is not != 15, result false
        int d = num1 | num2;             // 10 | 15 is 15, result: 15

        System.out.println(a);
        System.out.println(b);
        System.out.println(c);
        System.out.println(d);
    }
}
```

OUTPUT

```
true
10
false
15
PS C:\Users\gdean\Documents\2nd Year\Second Semester\COMP 009 Object Oriented Programming\Testing Codes Java>
```

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2. Create a Java program using the ternary operator that will produce the output: true, false, 25, 30.

CODE

Act2_Part2.java > Act2_Part2 > main(String[])

```java
public class Act2_Part2 {
        Run | Debug
        public static void main(String[] args) {
        int num1 = 20;
        int num2 = 35;

        boolean less_boolean = (num1 < num2) ? true : false;    // num1 is less than num2, result: true
        boolean greater_boolean = (num1 > num2) ? true : false; // num1 is not greater than num2, result: false
        int less_int = (num1 < num2) ? num1 : num2;             // num1 is less than num2, result: 20
        int greater_int = (num1 > num2) ? num1 : num2;          // num1 is not greater than num2, result: 35

        System.out.println(less_boolean);
        System.out.println(greater_boolean);
        System.out.println(less_int);
        System.out.println(greater_int);
    }
}
```

OUTPUT

```
true
false
20
35
PS C:\Users\gdean\Documents\2nd Year\Second Semester\COMP 009 Object Oriented Programming\Testing Codes Java>
```

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3. **Create a Java program using assignment operators that will produce the output: Addition: 120, Subtraction: 70, Multiplication: 144, Division: 16.**

CODE

Act2_Part3.java > ...

```java
public class Act2_Part3 {
    public static void main(String[] args) {

        int addition = 95;
        int subtraction = 95;
        int multiplication = 12;
        int division = 48;

        addition += 25;       // 95 + 25 = 120
        subtraction -= 25;    // 95 - 25 = 70
        multiplication *= 12; // 12 * 12 = 144
        division /= 3;        // 48 / 3 = 16

        System.out.println("Addition: " + addition);
        System.out.println("Subtraction: " + subtraction);
        System.out.println("Multiplication: " + multiplication);
        System.out.println("Division: " + division);
    }
}
```

OUTPUT

```
Addition: 120
Subtraction: 70
Multiplication: 144
Division: 16
PS C:\Users\gdean\Documents\2nd Year\Second Semester\COMP 009 Object Oriented Programming\Testing Codes Java>
```

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PART II:

Program Requirements:

1. **Use at least five (5) different operators from the following categories:**
   - **Arithmetic operators (+, -, *, /, %)**
   - **Relational operators (>, <, >=, <=,** ==, !=)
   - Logical operators (&&, ||, !)
   - Unary operators (++, --)
   - Assignment operators (+=, -=, *=, /=)
   - Bitwise operators (&, |, ^, <<, >>)

2. The program must produce exactly six (6) outputs only.
3. At least one output must include a tricky increment or decrement case, such as:
   - a++ + ++a
   - --b + b++
   - or any similar expression that modifies the variable within the same statement.
4. Do NOT reuse:
   - Any variable values
   - Any variable names
   - Any logic patterns
     from Part I.
5. The output must contain:
   - At least two (2) boolean results
   - At least four (4) numeric results

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CODE

```java
public class Act2_PartII{

    int num_result;
    boolean boolean_result;

    public void print_int(){
        System.out.println(num_result);
    }

    public void print_boolean(){
        System.out.println(boolean_result);
    }

    public static void main(String[] args){
        int num1 = 1200;
        int num2 = 35;

        Act2_PartII obj = new Act2_PartII();


        obj.num_result = num1 / 20;
        obj.print_int(); // Output: 60 (1200 / 20 = 60)


        obj.num_result = num2 *= 2;
        obj.print_int(); // Output: 70 (35 * 2 = 70)

        obj.num_result = (num2 != num1) ? 6 : 9;
        obj.print_int();
        // Output: 6 (ternary – num2 is 70, num1 is 1200, 70 != 1200 so returns 6)

        obj.num_result = (num1 > num2) ? 7 : 3;
        obj.print_int();
        // Output: 7 (ternary – num1 > num2 is true, so returns 7)

        obj.boolean_result = (num1 > 0) && (num2 > 0);
        obj.print_boolean();
        // Output: true (logical – num1 > 0 AND num2 > 0)

        obj.boolean_result = (num1 < num2);
        obj.print_boolean();
        // Output: false (num1 1200 is NOT > num2 1200... num1 > num2 is false since num2=70 < num1=1200)

    }

}
```

OUTPUT

```
60
70
6
7
true
false
PS C:\Users\gdean\Documents\2nd Year\Second Semester\COMP 009 Object Oriented Programming\Testing Codes Java>
```